

Use missing-number and input-output thinking

Step 4

 Kindergarten /
Early Elementary


Today I am learning to use missing-number and input-output thinking.

1. Find the Missing Number

Look at each pattern. Write the missing number.

a)

1	2	3		5	6
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b)

2	4		8	10	12
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c)

3	6	9		15	18
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d)

5	10	15	20		30
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2. What Comes Next?

Look at the patterns. Write the next number.

a) 10, 20, 30, 40, _____

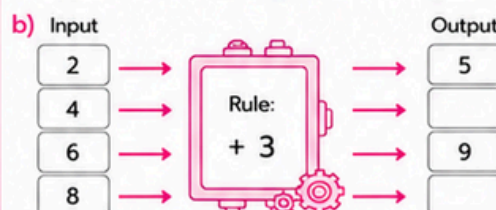
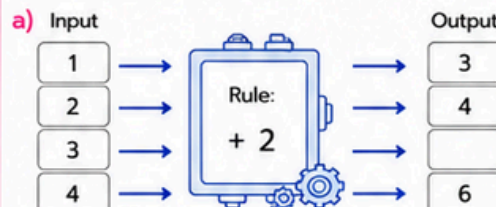
b) 2, 4, 6, 8, _____

c) 5, 10, 15, 20, _____

d) 3, 6, 9, 12, _____

3. Input → Output

Look at the rule in each machine. Write the missing output.



4. What Is the Rule?

Look at the input and output. What rule is used? Write the rule.

 a)

Input	1	2	3	4	5
Output	4	5	6	7	8

Rule: _____

 b)

Input	2	4	6	8	10
Output	6	8	10	12	14

Rule: _____

 c)

Input	5	7	9	11	13
Output	10	12	14	16	18

Rule: _____

5. Use the Rule

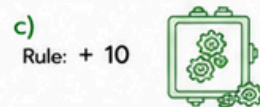
Use the rule to find the missing numbers.



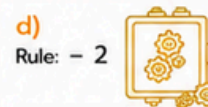
Input	1	2	3	4	5
Output	5				



Input	1	2	3	4	5
Output	6				



Input	1	2	3	4	5
Output	11				



Input	5	6	7	8	9
Output	3				

6. Solve the Problems

- a) There are 7 apples in a basket. 3 more apples are added. How many apples are there now?

 +  =

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- b) Sara has 9 crayons. She gives 4 crayons to her friend. How many crayons does Sara have left?

 -  =

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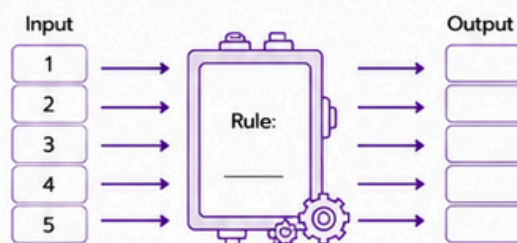
- c) There are 6 fish in a bowl. 4 more fish swim in. How many fish are in the bowl now?

 +  =

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7. Make Your Own Input-Output Machine

Choose a rule. Fill in the input and output numbers. Write your rule.



My rule is: _____

I can...



- ☐ find missing numbers in patterns.
- ☐ use input and output rules.
- ☐ solve problems using the rule.
- ☐ explain what is happening.



My Learning Goal

I can use missing-number and input-output thinking to solve problems.

